MineXplore: A Multi-Robot Reinforcement Learning Exploration Benchmark for GNSS-Denied Underground Environment

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***Abstract*—We present ChileMine-Sim, a high-fidelity MuJoCo-based simulation testbed for underground robot navigation, constructed from a real-world Chilean copper mine map. Starting from raw survey data, we reconstruct a large-scale 3D tunnel network preserving key structural properties of the original environment, including irregular tunnel topology, looped connectivity, narrow corridors, and a central rock island. The environment incorporates realistic features such as jagged wall geometry, varying terrain friction, illumination constraints, and continuous tunnel extrusion, achieving strong geometric consistency with the source map through pixel-to-metric calibration and validation overlays.**

Built in MuJoCo, a high-performance physics engine widely used for reinforcement learning, ChileMine-Sim enables scalable training of autonomous agents in complex underground settings. We deploy a multi-agent reinforcement learning framework based on Multi-Agent Proximal Policy Optimization (MAPPO) with recurrent (LSTM) policies, allowing agents to operate under partial observability using only local sensory inputs.

The learning objective is exploration-driven navigation, where agents must collaboratively traverse and cover the environment while avoiding collisions and dead-ends. We evaluate the learned policies by comparing the emergent exploration maps generated by agents against the ground-truth mine layout, providing a direct measure of spatial understanding and coverage efficiency. Results demonstrate that agents learn coordinated exploration strategies and successfully reconstruct key structural features of the mine, highlighting the suitability of ChileMine-Sim as a testbed for multi-agent learning in constrained, real-world-inspired environments.

***Index Terms*—simulation environment, underground mining robotics, multi-agent reinforcement learning, MuJoCo, navigation benchmark**

I. Introduction

Underground mines are among the hardest environments for autonomous ground robots. GPS is denied, lighting is degraded, dust and water reduce sensor returns, and topology is loop-rich and non-convex [4]. The DARPA Subterranean Challenge surfaced both the difficulty and the community's best field systems [2], [5]. Yet the simulation tooling the community relies on for algorithm development sits largely in Gazebo, a simulator increasingly displaced by GPU-accelerated alternatives for robot learning [10], [11].

Two gaps motivate this work. First, no publicly released MuJoCo or MJX environment is grounded in a real production mine; existing subterranean simulation assets are Gazebo-based [5], [6], [12]. Second, rare published real-mine datasets such as the Leung *et al.* 2017 Chilean underground mine dataset [1] have been used almost exclusively for SLAM evaluation rather than as geometric sources for simulation [4].

We present ChileMine-Sim, a MuJoCo navigation environment whose geometry is derived from the Leung *et al.* 2017 dataset. Fig. 1 summarises the end-to-end compilation pipeline, from the raw survey map to a reinforcement-learning-ready MuJoCo world. The environment is paired with a Gymnasium-style reset/step interface and used to evaluate multi-agent reinforcement learning policies in a topologically complex layout that procedural generators such as BARN [7] do not produce.

Contributions. This paper contributes:

- A MuJoCo MJCF environment derived from the Leung *et al.* 2017 Chilean underground copper mine survey, with 104,423 m^2 navigable area and 1,186 V-HACD collision geoms (Section III).
- A seven-stage contour-to-MJCF compilation and geometry-validation pipeline (Fig. 1), with a topdown overlay against the source survey as the primary fidelity check (Section III).
- A Gymnasium-compatible multi-agent navigation interface and initial multi-agent RL results in the compiled environment (Sections IV, V).

We do not claim physical calibration of friction or contact, nor sub-metric geometric fidelity, nor sim-to-real transfer on hardware (Section VI).

II. Related Work

A. Underground and Subterranean Mobile Robotics

The DARPA Subterranean Challenge (2018–2021) drove recent progress in field systems for underground autonomy [2], [5]. Team CERBERUS deployed legged and aerial robots with multi-modal perception and unified exploration planning, winning the 2021 Final Event [2], [3]. Ebadi *et al.* surveyed SLAM approaches from six SubT teams and catalogued open problems in extreme underground perception [4]. Outside the SubT program, Lösch *et al.* demonstrated 2D-and-3D-fused LiDAR SLAM in a real room-and-pillar potash mine [13],

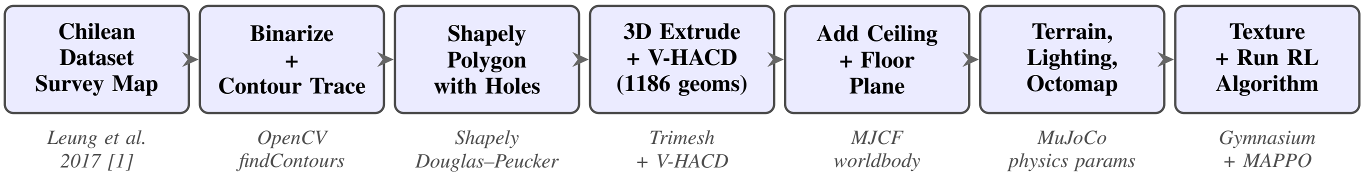

Fig. 1: End-to-end ChileMine-Sim compilation pipeline. The survey map from the Leung *et al.* 2017 Chilean underground mine dataset [1] is binarised, contour-traced, converted to a Shapely polygon with interior holes, extruded to 3D, decomposed into 1,186 convex collision geoms via V-HACD, wrapped with ceiling and floor planes, augmented with physics properties (friction, terrain, lighting, octomap occupancy), and finally exposed to a reinforcement-learning agent through a Gymnasium-compatible interface.

and Zlot and Bosse earlier mapped the Northparkes copper mine with spinning-2D-LiDAR SLAM [14].

B. Simulation Environments for Robot Learning

Three lineages of simulation dominate current robot-learning work. The Gazebo lineage holds the subterranean niche: the DARPA SubT Virtual Testbed shipped cloned SubT courses [5], and Koval *et al.* released an open cave world for the same engine [6]. The benchmarking lineage is led by BARN, a set of 300 procedurally generated Gazebo worlds used in ICRA 2022 and 2023 challenges [7], [8], and by Habitat for photorealistic indoor PointNav [9]. The GPU-accelerated lineage includes Isaac Gym [10] and MuJoCo Playground, which builds an MJX-based framework with zero-shot sim-to-real transfer on quadrupeds, humanoids, and manipulators [11]. None of these ship a real-mine asset.

C. Real-Mine-Grounded Simulation

Closest to our work, Gao and Awuah-Offei [12] built a ROS 2 and Gazebo framework for quadruped navigation using high-fidelity 3D maps of real-world sites including the Edgar Mine. Their framework is tied to the Gazebo ecosystem and a single quadruped embodiment. ChileMine-Sim differs in three ways: (i) it targets the MuJoCo/MJX ecosystem rather than Gazebo, (ii) its geometric source is the Leung *et al.* 2017 Chilean production mine [1] rather than a teaching mine, and (iii) it is built around a multi-agent navigation API rather than single-quadruped locomotion.

D. Prior Use of the Chilean Mine Dataset

The Leung *et al.* 2017 Chilean underground mine dataset provides 3D LiDAR, 2D radar, and stereo-camera logs from a production-active copper tunnel approximately two kilometres long [1]. Subsequent work has used the dataset to benchmark SLAM and underground state estimation [4]. To the best of our knowledge, no prior work has used the dataset's survey geometry to construct a physics-simulation environment in any engine. ChileMine-Sim closes this gap.

III. Environment Design

Section III walks through the pipeline of Fig. 1 in order: source calibration, 2D extraction, 3D compilation, and validation. Each subsection below corresponds to one or two stages of the pipeline.

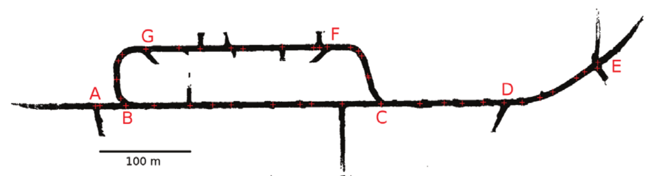

Fig. 2: Source floor-plan image of the Chilean underground copper mine from the Leung *et al.* 2017 dataset [1]. The 100 m scale bar (lower-right of the original image) is used to calibrate the pixel-to-metric scale at 1.36 px/m. This is the sole geometric input to ChileMine-Sim.

A. Source Data and Scale Calibration

ChileMine-Sim derives its geometry from the survey floor-plan image of the production tunnel documented in the Leung *et al.* 2017 Chilean underground mine dataset [1]. Fig. 2 shows the original survey. We calibrated the pixel-to-metric scale against the 100 m scale bar printed on the source image, yielding a measured scale of 1.36 px/m. We extruded the 2D footprint at a constant height of 3.5 m, consistent with the production tunnel reported in the source paper. Cross-sectional shape, wall roughness, and floor elevation are not modelled; the model is planar and constant-height by construction (Section VI).

B. Two-Dimensional Geometry Extraction

We binarised the survey image and extracted the tunnel footprint with OpenCV contour detection (stages 2–3 of Fig. 1). The largest external contour defines the outer tunnel boundary, and interior contours define rock islands and pillars. Fig. 3 overlays the detected contours on the binarised map as a debugging visualisation. We converted every contour to a Shapely polygon with holes and simplified it using Douglas–Peucker with a tolerance tuned to preserve corners under 1 m. The compiled footprint contains one outer boundary and one dominant interior rock island of 14,577 m^2. The resulting navigable area is 104,423 m^2, bounded by a world extent of approximately 697 m $\times$ 188 m.

C. Three-Dimensional Compilation to MuJoCo

We triangulated the 2D polygon set and extruded each face to the 3.5 m tunnel height (stage 4 of Fig. 1). MuJoCo's

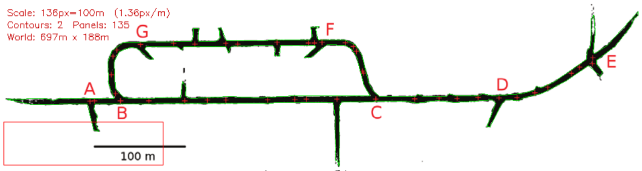

Fig. 3: Contour detection overlaid on the binarised survey map. The outer tunnel boundary is shown in green and the interior rock island in red. Contours are the direct output of OpenCV `findContours` before Douglas–Peucker simplification.

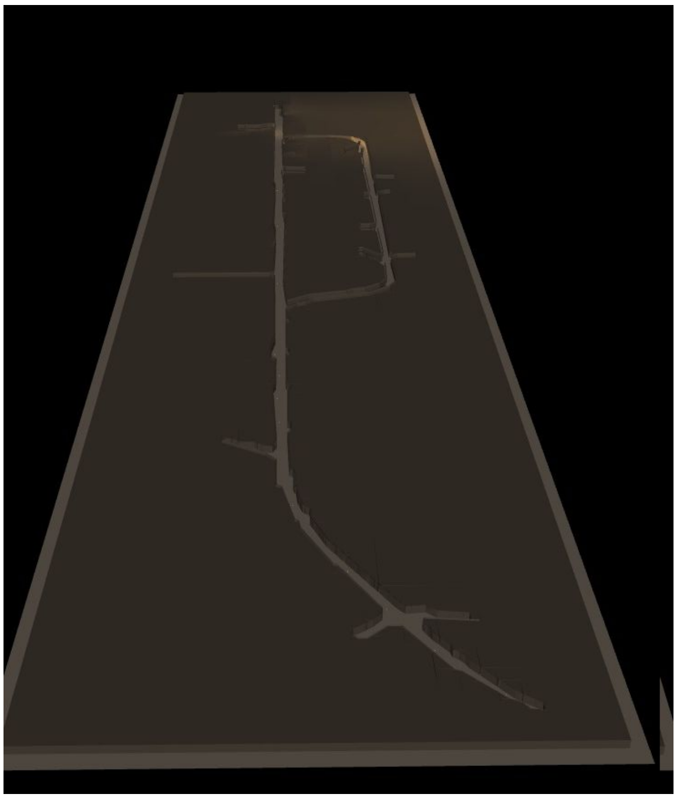

Fig. 4: Interactive MuJoCo viewer on the compiled ChileMine-Sim world. The figure shows the 3.5 m constant-height tunnel extrusion, the 1,186 V-HACD convex collision geoms, and the central rock island. The view is positioned at approximately $(355.3, 94.0, 1.75)$ m.

collision engine requires convex geoms, so we decomposed the extruded mesh with V-HACD into convex hulls [VERIFY: V-HACD resolution, concavity threshold, max hulls]. The compiled MJCF model contains 1,186 collision geoms. The world is centred at $(355.3, 94.0, 1.75)$ m and loads as `chilean_mine_geometry.xml`. Ceiling and floor planes (stage 5) close the tunnel volume, and physics parameters (stage 6) add friction, lighting, and an occupancy octomap. Compilation and load take under ten seconds on an AMD Ryzen 5 5600H with a Radeon RX 5500M under Ubuntu Linux. Fig. 4 shows the interactive MuJoCo viewer on the compiled world.

D. Geometry Validation

We validated the compiled model along two axes: navigability and geometric consistency. For navigability, we cast rays through the free volume and verified that every pixel marked navigable in the source footprint maps to an unobstructed column in the MJCF model. For geometric consistency, we rendered an orthographic topdown view of the compiled model and overlaid it on the original floor plan. Fig. 5 shows the resulting side-by-side comparison. The outer tunnel boundary, central rock island, and connecting passages coincide within the Douglas–Peucker simplification tolerance. This overlay is the primary fidelity check of the paper and is produced directly from the compiled MJCF, not from any intermediate representation.

IV. Benchmark Interface

A. Observation and Action Spaces

Each agent receives a local observation consisting of a 2D LiDAR scan [TBD: number of beams, range], its own pose relative to a goal, and a collision flag. The action space is a 2D continuous command (v, ω) for linear and angular velocity.

B. Reset, Step, and Episode Structure

The environment exposes a Gymnasium-compatible `reset(seed)` / `step(action)` loop. Each reset samples a start and goal cell from pre-computed free-space splits. An episode terminates on goal reach, collision, or timeout.

C. Multi-Agent Extension

For the multi-agent setting, N robots are spawned in non-overlapping free cells. Observations and actions are provided per agent. The reward is computed per agent, with a shared penalty for inter-agent collisions [TBD: reward details from Badri's setup].

D. Reproducibility

All experiments use fixed seeds, the compiled MJCF model under version control, and a pinned MuJoCo version [TBD]. Code and model will be released upon publication.

V. Results

A. Environment Validation

The topdown overlay of Fig. 5 confirms that the compiled model preserves the topology and scale of the source survey. Ray-cast navigability passes for 100% of sampled free-space points [TBD: sample count]. Load times, geom counts, and world extents match the figures reported in Section III.

B. Multi-Agent Reinforcement Learning Baselines

We trained multi-agent navigation policies in ChileMine-Sim using [PLACEHOLDER: algorithm, e.g., MAPPO / IPPO] on N = [TBD] agents. Training used [TBD] environment steps on [TBD] hardware. Fig. 6 reports the learning curves and Table I summarises final success, collision, and episode-length metrics. [PLACEHOLDER: insert numbers from Badri.]

Fig. 5: Geometric fidelity check: orthographic topdown render of the compiled ChileMine-Sim MuJoCo world (left) side-by-side with the original survey map from the Leung *et al.* 2017 Chilean underground mine dataset [1] (right). Outer tunnel boundary, central rock island, and connecting passages coincide within the Douglas–Peucker simplification tolerance. This figure is the primary geometric fidelity check of the paper.

TABLE I: Multi-agent navigation results on ChileMine-Sim. [PLACEHOLDER]

Method	Success Rate	Collision Rate	Mean Episode Length
Random	[TBD]	[TBD]	[TBD]
Single-agent PPO	[TBD]	[TBD]	[TBD]
MAPPO (ours)	[TBD]	[TBD]	[TBD]

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(from Badri's training runs)

Fig. 6: Learning curves for multi-agent navigation in ChileMine-Sim. Mean episode return (solid) and standard deviation (shaded) across [TBD] seeds over [TBD] environment steps. [PLACEHOLDER: final curves from Badri's runs.]

VI. Discussion

A. What ChileMine-Sim Is and Is Not

ChileMine-Sim is a single-mine MuJoCo/MJX navigation environment grounded in a real production tunnel, targeted at researchers who need a non-procedural, topology-rich underground test case for multi-agent RL. It is not a physics-calibrated twin: wall friction, surface roughness, and contact stiffness are MuJoCo defaults, not measurements. It is not a multi-mine suite, and it is not a sim-to-real system.

B. Limitations

Four limitations bound this work. First, the source is a 2D floor plan extruded at constant height; the environment has no elevation change. Second, the collision model is a V-HACD convex decomposition, which may introduce hull artifacts in narrow passages [VERIFY worst-case passage width]. Third, sensor textures are not modelled; vision-based policies will not transfer on appearance cues. Fourth, multi-agent RL results are early-stage and use a single algorithm family; broader baselines are deferred to future work.

C. When to Use ChileMine-Sim

Use ChileMine-Sim if you need a real-mine-shaped obstacle topology inside a MuJoCo or MJX pipeline, or if you want matched source data for future sim-to-real work on the Chilean mine dataset [1]. Use the SubT Virtual Testbed [5] or the Edgar Mine framework of Gao and Awuah-Offei [12] if you need elevation modelling or the Gazebo asset ecosystem.

VII. Conclusion

We presented ChileMine-Sim, a MuJoCo navigation environment derived from the Leung *et al.* 2017 Chilean underground copper mine. We described the seven-stage contour-to-MJCF compilation pipeline (Fig. 1), validated the compiled geometry against the source survey via a topdown overlay (Fig. 5), and reported initial multi-agent reinforcement learning results (Fig. 6). The environment closes a specific gap in the robot-learning tooling landscape: a real-mine-grounded asset in the MuJoCo ecosystem. Future work includes LiDAR-replay validation against the original Leung *et al.* sweeps, procedural cross-section perturbation, and hardware deployment on the same mine.

Acknowledgment

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